

Orbit Determination for Impulsively Maneuvering Spacecraft Using Modified State Transition Tensor

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This paper proposes a method to accurately resolve orbit determination for a spacecraft with unknown impulsive maneuvers. The proposed method handles the unknown impulsive maneuver by incorporating the magnitude, direction, and time of the impulsive maneuver into the estimation parameter vector. First, a modified state transition tensor (STT) is proposed via orbit division and segment connection, allowing the orbit to be directly propagated under the effects of impulsive maneuver uncertainties. Then, based on the modified STT, a second-order measurement model is established with the estimation parameter vector as the input. Combining the second-order measurement model with observations, a second-order optimal solution is derived to correct the estimation parameters. The spacecraft orbit, together with the magnitude, direction, and time of the impulsive maneuver, are simultaneously estimated in an iterative framework. The performance of the proposed method is validated in a low-Earth-orbit case and a high-Earth-orbit case. Simulations show that the proposed method outperforms its linear version in terms of convergence, accuracy, and uncertainty quantification capacity. Its maneuver reconstruction and orbit estimation errors are one order of magnitude less than those of competitive methods. Moreover, the proposed method can handle severe conditions and is robust to initial guesses.

I. Introduction

ACCURATE tracking, estimation, and prediction of a spacecraft orbit are critically important in all space missions. Several estimators, for example, the least-squared method (LSM) and its variants [1] and the Kalman filter (KF) and its variants [2–4], have been widely used in solving the orbit determination (OD) problem. One of the complicated and challenging tasks in OD is to determine and predict the orbit of a maneuvering spacecraft, especially a noncooperative target with unknown maneuvers [5,6]. Without correctly compensating for unknown maneuvers, the OD performance of conventional estimators is significantly degraded, and sometimes these methods even fail to converge [7,8].

The state-of-the-art means of estimating the orbit of a maneuvering spacecraft can be broadly divided into three categories: reinitiation of the OD process, the filtering-through approach, and maneuver reconstruction [7]. Reinitiation methods estimate the postmaneuver orbit by directly inflating the estimation covariance [9], disregarding the premaneuver measurements and OD solutions. Unknown maneuvers are compensated by the largely inflated initial uncertainty during the state update stage. Reinitiation methods are simple and easy to operate. However, their performance relies on precise maneuver detection and careful selection of the inflating weights. Thus, they are not satisfactory for scenarios with high precision requirements [10].

The filtering-through approach is a real-time tracking technique compensating for unknown maneuvers to fit the postmaneuver observation into the premaneuver orbit [7,11]. According to the methods

for compensating the unknown maneuver, the filtering-through approach developed thus far can be further cataloged into three groups: the single-model-based adaptive KFs, multiple-model algorithms, and decision-based adaptive KFs [12]. The single-model-based adaptive KFs compensate for the unknown maneuvering accelerations using a single model, including the thrust-Fourier-coefficient (TFC) event representation [7,13,14], the polynomial representation [6], and the variable structure estimator [15]. The multiple-model algorithms use a bank of Kalman filters with different maneuvering submodels in parallel, in which each submodel corresponds to one particular type of maneuver. Typical multiple-model algorithms include the interacting multiple model (IMM) algorithm and its variants [16,17]. The disadvantage of IMM algorithms is that they only work well when the submodels accurately fit the actual maneuvering acceleration. Moreover, the IMM algorithms suffer from algorithm complexity as their computational load increases linearly with the number of submodels [18,19]. The decision-based adaptive KF catalogs include the equivalent noise approaches [20], the input estimation (IE) approaches [21,22], and the switching model (SM) approaches [23,24]. The equivalent noise approaches and the IE approaches can only be applied to address certain types of maneuvers, and their performances significantly degrade when the actual maneuver varies with time [25]. The SM approaches use nonmaneuvering and maneuvering dynamics and switch according to the maneuver detection results. The SM approaches can perform well but strongly rely on high-frequency observations and accurate maneuver detection [15]. A major concern for the filtering-through approaches is that, although compensating the unknown maneuvers can help improve the estimation accuracy, it is often expected to obtain more information about the maneuver to better track the target, that is, to reconstruct the maneuver [26,27].

Maneuver reconstruction methods are unified by their characterization in postprocessing. There are many ways to reconstruct the unknown maneuver, but all are based on independent estimations of the pre- and postmaneuver orbits [27–29]. The weakness of such methods is that they are sensitive to the estimation errors of the pre- and postmaneuver orbits [29]. In all current maneuver reconstruction methods, the reconstruction process is independent of the orbit estimation. Orbit estimation errors are inevitable due to the measurement noise and the scarce knowledge of the maneuver. In turn, the orbit estimation errors couple and amplify into the errors of the

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maneuver reconstruction solutions [27,30]. In particular, it is found that the maneuver time is most likely to be distorted by position uncertainties [27]. Moreover, compared with the first two categories, the maneuver reconstruction method usually has heavy computational overhead [7].

The previously mentioned methods are mainly developed for ideal OD scenarios, assuming all data preprocessing is done. There is also some literature that contributes to solving more difficult and complex versions of maneuvering spacecraft OD problems. Holzinger et al. proposed an optimal control-based estimation (OCBE) method for multimaneuvering target tracking [31,32]. The OCBE uses a control effort as a defined metric to correlate target observations, detect maneuvers, and characterize maneuvers before estimating the orbit. Hall and Singla developed a unified reachability-based framework for tracking lost-in-space noncooperative maneuvering targets [33,34]. Their method uses numerical quadrature methods to efficiently propagate the reachability sets of the maneuvering target [35,36], which can be further employed for searching lost-in-space targets and tasking sensors, maneuver detection and estimation, and orbit estimation [37].

The method presented in this paper falls into the maneuver reconstruction category. The key difficulty of the impulsively maneuvering spacecraft OD lies in accurately estimating the vector and the time of the unknown impulsive maneuver. One single unknown impulsive maneuver is assumed to occur to simplify the problem. The proposed method incorporates the vector and the time of the unknown impulsive maneuver into the estimation parameter vector when modeling the OD system. It can estimate the orbits and reconstruct the impulsive maneuver simultaneously. As such, the effects of the orbit estimation errors on the maneuver reconstruction are reduced, while the accuracy and robustness of the maneuver reconstruction method are improved. This paper first models the effects of the impulsive maneuver vector and epoch on the orbits from the view of uncertainty propagation. Under the effects of the impulsive maneuver, the orbit of the spacecraft is divided into several segments. A virtual state is defined to handle the time uncertainty of the impulsive maneuver. Based on the original state transition tensor (STT) technique, a modified second-order STT is proposed for the orbit propagation considering impulsive maneuver uncertainties via virtual state reversion and segment connection. Then, based on the modified STT, a second-order measurement model is proposed to establish the mapping relationship from the estimation parameter vector to the measurements. Starting from the given a priori parameter estimation, the reference orbit is propagated using the modified STT, and the residuals between incoming observation data and predicted measurements are calculated. Based on the partials provided by the second-order measurement model, the concept of the second-order optimal estimation is extended from the conventional least-squares solution. The correction to the a priori parameter estimation is made from the second-order optimal estimation. The impulsively maneuvering spacecraft OD problem is solved in an iterative framework, and thus the algorithm efficiency is greatly improved.

The remainder of this paper is organized as follows. Section II describes the impulsively maneuvering spacecraft OD problem. Section III addresses the impulsively maneuvering spacecraft OD problem using the modified STT technique. The second-order measurement model and the second-order optimal estimation solution are presented in detail. In Sec. IV, numerical simulations are performed to investigate the performance of the proposed algorithm. Finally, conclusions are given in Sec. V.

II. Preliminaries

A. Description of Impulsively Maneuvering Spacecraft OD Problem

Consider an OD problem in which the target executes one unknown impulsive maneuver. Assume that the orbit state of the target is labeled as $\mathbf{x} = [\mathbf{r}; \mathbf{v}]$, where $\mathbf{r} = [x, y, z]^T$ and $\mathbf{v} = [\dot{x}, \dot{y}, \dot{z}]^T$ denote the position and velocity vectors in the Earth-centered equatorial inertial (ECEF) coordinate system, respectively. The OD process begins at epoch t_0 and ends at epoch t_2 . An impulsive maneuver

$\Delta \mathbf{v}_1 = [\Delta v_x; \Delta v_y; \Delta v_z]$ is applied at epoch t_1 , where $t_1 \in (t_0, t_2)$. The conventional OD process only estimates the orbit of the spacecraft. In this paper, we aim to estimate the orbit and the time and vector of the impulsive maneuver together. Thus, the estimation parameter vector $\mathbf{X}$ is written as

$$\mathbf{X} = [\mathbf{r}; \mathbf{v}; \Delta \mathbf{v}_1; t_1] \quad (1)$$

The system equation of the impulsively maneuvering spacecraft OD problem is given by

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{y}(\mathbf{x}, t) \in \mathbb{R}^6 \\ \mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{e} \in \mathbb{R}^d \end{cases} \quad (2)$$

where $\mathbf{y}(\mathbf{x}, t) \in \mathbb{R}^6$ represents the state model, $\mathbf{h}(\mathbf{x}) \in \mathbb{R}^d$ is the measurement model, d is the dimension of the measurement vector, and $\mathbf{e} \in \mathbb{R}^d$ is the Gaussian-distributed measurement noise with the covariance matrix represented by $\mathbf{R} \in \mathbb{R}^{d \times d}$.

Finally, the point mass gravity of Earth and the Earth's oblateness perturbation J_2 are expressed as

$$\begin{cases} \dot{\mathbf{r}} = \mathbf{v} \\ \dot{\mathbf{v}} = -\frac{\mu_e}{\|\mathbf{r}\|^3} \mathbf{r} + \mathbf{a}_{J_2}(\mathbf{r}) \end{cases} \quad (3)$$

where $\mu_e = 398600.44 \text{ km}^3/\text{s}^2$ is the gravitational constant of the Earth and $\mathbf{a}_{J_2}(\mathbf{r})$ is the perturbation acceleration of J_2 , given by

$$\mathbf{a}_{J_2}(\mathbf{r}) = \begin{bmatrix} \mu_e x J_2 \frac{R_e^2}{\|\mathbf{r}\|^5} \left(-\frac{3}{2} + \frac{15}{2} \frac{z^2}{\|\mathbf{r}\|^2} \right) \\ \mu_e y J_2 \frac{R_e^2}{\|\mathbf{r}\|^5} \left(-\frac{3}{2} + \frac{15}{2} \frac{z^2}{\|\mathbf{r}\|^2} \right) \\ \mu_e z J_2 \frac{R_e^2}{\|\mathbf{r}\|^5} \left(-\frac{9}{2} + \frac{15}{2} \frac{z^2}{\|\mathbf{r}\|^2} \right) \end{bmatrix} \quad (4)$$

where $R_e = 6378.137 \text{ km}$ denotes the Earth's radius and $J_2 = 1.08264 \times 10^{-3}$.

B. State Transition Tensor Technique

The STT technique is a classic analytical nonlinear propagation method derived from the higher-order Taylor series approach [38–41]. Let $\mathbf{x}_0$ and $\mathbf{x}(t)$ denote the state at the initial epoch t_0 and an arbitrary future epoch t , respectively. From Eq. (2), the deviation of the state $\mathbf{x}$ from a reference (nominal) orbit can be approximated by the N th order Taylor series expansion at the initial state $\mathbf{x}_0$ as

$$\delta \mathbf{x}^i(t) = \sum_{p=1}^N \frac{1}{p!} \Phi_{(t_0,t)}^{i,k_1 \dots k_p} \delta x_0^{k_1} \dots \delta x_0^{k_p} \quad (5)$$

where $x^i(t)$ ($i \in \{1, 2, 3, 4, 5, 6\}$) denotes the i th elements of the state vector $\mathbf{x}(t)$, $x_0^{k_j}$ ($k_j \in \{1, 2, 3, 4, 5, 6\}$) is the k_j th component of the state vector $\mathbf{x}_0$, and $\Phi_{(t_0,t)}^{i,k_1 \dots k_p}$ is the p th order STT from t_0 to t , given by

$$\Phi_{(t_0,t)}^{i,k_1 \dots k_p} = \frac{\partial^p x^i(t)}{\partial x_0^{k_1} \dots \partial x_0^{k_p}} \quad (6)$$

Note that in Eqs. (5) and (6) (and the equations in the following parts of this paper), the Einstein summation notation is used for all the dummy variables (i.e., $k_1, k_2, \dots, k_p$) [42,43]. For example, for the second-order term $\Phi_{(t_0,t)}^{i,k_1 k_2} \delta x_0^{k_1} \delta x_0^{k_2}$, the subscripts k_1 and k_2 traverse from 1 to 6 in the summation, such that

$$\Phi_{(t_0,t)}^{i,k_1 k_2} \delta x_0^{k_1} \delta x_0^{k_2} = \sum_{k_1=1}^6 \sum_{k_2=1}^6 \Phi_{(t_0,t)}^{i,k_1 k_2} \delta x_0^{k_1} \delta x_0^{k_2} \quad (7)$$

The STTs $\Phi_{(t_0,t)}^{i,k_1 \dots k_p}$ can be obtained by integrating the differential equations. The differential equations for obtaining the STTs (up to second order) are expressed as

$$\dot{\Phi}_{(t_0,t)}^{i,k_1} = y_{\text{ref}}^{i,p_1} \Phi_{(t_0,t)}^{p_1,k_1} \quad (8)$$

$$\dot{\Phi}_{(t_0,t)}^{i,k_1 k_2} = y_{\text{ref}}^{i,p_1} \Phi_{(t_0,t)}^{p_1,k_1 k_2} + y_{\text{ref}}^{i,p_1 p_2} \Phi_{(t_0,t)}^{p_1,k_1} \Phi_{(t_0,t)}^{p_2,k_2} \quad (9)$$

where p_1 and p_2 are dummy variables used for summation convention; $y_{\text{ref}}^{i,k_1 \dots k_p}$ is the partials of the dynamic equation $y(x, t)$ computed along the reference orbit x_{ref} , given by

$$y_{\text{ref}}^{i,k_1 \dots k_p} = \left. \frac{\partial^p y^i(x, t)}{\partial x^{k_1} \dots \partial x^{k_p}} \right|_{x=x_{\text{ref}}} \quad (10)$$

III. Methodology

This section divides the proposed method of solving the impulsively maneuvering spacecraft OD problem into three parts. First, the modified STT for orbit propagation is proposed, considering the impulsive maneuver vector and epoch uncertainty. Then, the second-order measurement model is derived based on the modified STT. The second-order optimal estimation is proposed by combining the second-order measurement model and the incoming observations, and a multiple-iterative solving framework is presented. Finally, the overall procedures of the proposed methodology are summarized.

A. Modified State Transition Tensor Considering Impulsive Maneuver Uncertainty

Assume that the target executes one unknown impulsive maneuver during $[t_0, t_2]$. The impulsively maneuvering spacecraft OD problem is shown in Fig. 1. The states of the real orbit and the reference orbit are labeled by $\tilde{x}(t)$ and $x(t)$, respectively. The blue lines represent the reference state [$x_0 = x(t_0)$, x_1^- , x_1^+ and $x_2 = x(t_2)$], while the red lines represent the real state [$\tilde{x}_0 = \tilde{x}(t_0)$, $\tilde{x}_1^-$, $\tilde{x}_1^+$ and $\tilde{x}_2 = \tilde{x}(t_2)$]. The green state x_1^* is named the virtual state, which is used as an intermediate variable to assist the formula derivation in this paper.

As shown in Figs. 1a and 1b, the definitions of the virtual state x_1^* are different in the cases of $\delta t_1 < 0$ and $\delta t_1 \geq 0$. For the case of $\delta t_1 < 0$, the virtual state is defined as the state of $\tilde{x}(t)$ at epoch t_1 (see Fig. 1a). For the case of $\delta t_1 \geq 0$, x_1^* is the state obtained by propagating $\tilde{x}_1^+$ back from epoch $\tilde{t}_1$ to t_1 (see Fig. 1b). Note that for the two considered cases the proposed modified STTs are the same. The reference state can be propagated according to the current estimations in a LSM or a KF process; x_0 and x_2 are the state at the beginning epoch and the final epoch in the OD process, respectively; x_1^- and x_1^+ are the states before and after the instance of the impulsive maneuver; $\Delta x_1 = x_1^+ - x_1^- = [0_{3 \times 1}; \Delta v_1]$ is the impulsive maneuver; δx_0 , δx_1^- , δx_1^+ , $\delta \Delta x_1 = [0_{3 \times 1}; \delta \Delta v_1]$, δx_2 , and δt_1 are the deviation states.

Assuming that the dynamics are accurate, the orbit during $[t_0, t_2]$ can be fully determined by x_0 , t_1 , and Δx_1 (Δv_1). With the current estimations x_0 , t_1 , and Δv_1 obtained, the goal of the impulsive OD is to determine the deviations δx_0 , $\delta \Delta x_1 = [0_{3 \times 1}; \delta \Delta v_1]$, and δt_1 .

The modified STT is used to analytically model the relationship between the deviation $\delta x(t)$ ($t_0 \leq t \leq t_2$) and the deviations δx_0 , $\delta \Delta x_1$, and δt_1 . According to the definitions of epochs t_0 , t_1 , $\tilde{t}_1$, and t_2 , in this paper, the orbit of the impulsively maneuvering spacecraft during $[t_0, t_2]$ is divided into three segments. The first orbit segment begins from the epoch t_0 and ends at the epoch $\tilde{t}_1$. The second segment is between epochs $\tilde{t}_1$ and t_1 . The third segment is in $[t_1, t_2]$. The modified STT is proposed by connecting these segments.

Consider the first segment. In the period $t_0 \leq t \leq t_1$, the deviation state $\delta x(t)$ can be represented by a second-order expression as

$$\delta x^k(t) = \Phi_{(t_0,t)}^{k,p} \delta x_0^p + \frac{1}{2} \Psi_{(t_0,t)}^{k,p_1 p_2} \delta x_0^{p_1} \delta x_0^{p_2} \quad (11)$$

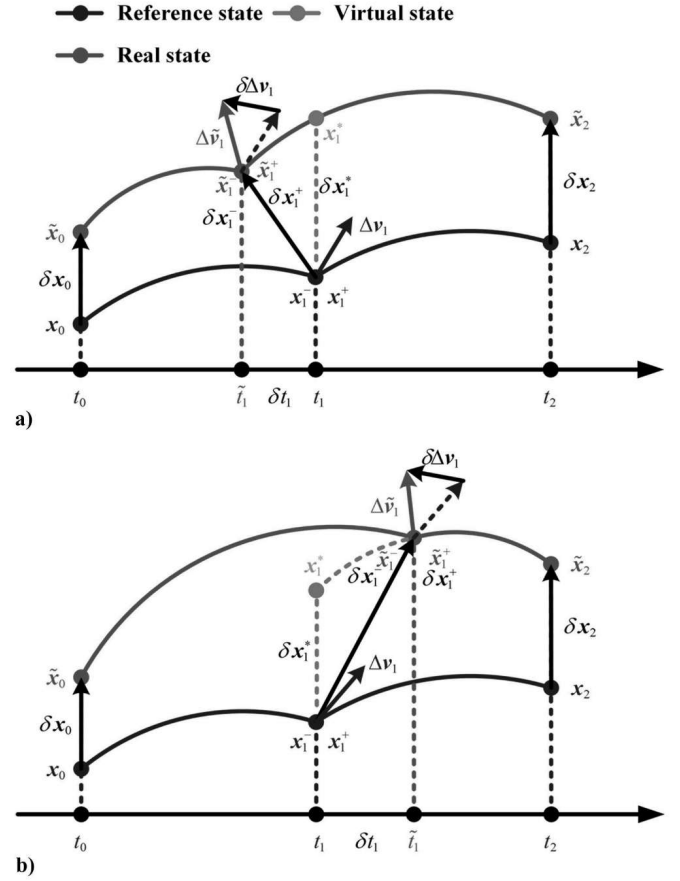


Fig. 1 A description of the variables in an impulsive OD problem: a) the case of $\delta t_1 < 0$ and b) the case of $\delta t_1 \geq 0$.

where $\Phi_{(t_0,t)}^{k,p}$ and $\Psi_{(t_0,t)}^{k,p_1 p_2}$ are the first-order STT and second-order STT of the dynamics in Eq. (3), $\delta x^k(t)$ is the k th element of the $\delta x(t)$, and δx_0^p is the p th element of δx_0 . We use the second-order expression in this study because the second-order STTs can adequately capture the nonlinearity of the Earth-centered dynamics [42,44].

Substituting $t = t_1$ into Eq. (11), we have

$$\delta x^k(t_1) = \Phi_{(t_0,t_1)}^{k,p} \delta x_0^p + \frac{1}{2} \Psi_{(t_0,t_1)}^{k,p_1 p_2} \delta x_0^{p_1} \delta x_0^{p_2} \quad (12)$$

Then, taking the influence of δt_1 into consideration, the second-order solution to the deviation state δx_1^- is expressed as

$$\begin{aligned} \delta x_1^{-k} &= \delta x^k(t_1) + f_-^k \delta t_1 + \frac{1}{2} g_-^k \delta t_1 \delta t_1 + H_-^{k,p} \delta t_1 \delta x_0^p \\ &= \Phi_{(t_0,t_1)}^{k,p} \delta x_0^p + \frac{1}{2} \Psi_{(t_0,t_1)}^{k,p_1 p_2} \delta x_0^{p_1} \delta x_0^{p_2} + f_-^k \delta t_1 \\ &\quad + \frac{1}{2} g_-^k \delta t_1 \delta t_1 + H_-^{k,p} \delta t_1 \delta x_0^p \end{aligned} \quad (13)$$

where δx_1^{-k} is the k th component of the state vector δx_1^- , and

$$f_-^k = \frac{\partial x_1^{-k}}{\partial t_1} = y^k(x_1^-, t_1) \quad (14)$$

$$g_-^k = \frac{\partial^2 x_1^{-k}}{\partial t_1 \partial t_1} = \frac{\partial y^k(x_1^-, t_1)}{\partial t_1} \quad (15)$$

$$\begin{aligned} H_-^{k,p} &= \frac{\partial^2 x_1^{-k}}{\partial t_1 \partial x_0^p} = \frac{\partial}{\partial t_1} \left(\frac{\partial x_1^{-k}}{\partial x_0^p} \right) = \frac{\partial [\Phi_{(t_0,t_1)}^{k,p}(x_1^-)]}{\partial t_1} \\ &= \dot{\Phi}_{(t_0,t_1)}^{k,p}(x_1^-) = \frac{\partial y^k(x_1^-, t_1)}{\partial x_1^{-p}} \end{aligned} \quad (16)$$

Equations (13–16) are the proposed second-order propagators of the first segment.

Now, we investigate the propagation in the second segment. By adding the deviation of the impulsive maneuver $\delta\Delta\mathbf{x}_1$ into Eq. (13), the deviation state $\delta\mathbf{x}_1^+$ is expressed as

$$\delta\mathbf{x}_1^{+,k} = \delta\mathbf{x}_1^{-,k} + \delta\Delta\mathbf{x}_1^k \quad (17)$$

According to the definition of the virtual state $\mathbf{x}_1^*$, the relationship between $\delta\mathbf{x}_1^*$ and $\delta\mathbf{x}_1^+$ can be found by performing a Taylor series expansion in terms of the state $\mathbf{x}_1^+$, given by

$$\begin{aligned} \delta\mathbf{x}^i(t) = & \underbrace{\Phi_{(t_1,t)}^{i,k} A_1^{k,p} \delta x_0^p + \frac{1}{2} \left(\Phi_{(t_1,t)}^{i,k} A_2^{k,p_1 p_2} + \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,p_1} A_1^{k_2,p_2} \right) \delta x_0^{p_1} \delta x_0^{p_2}}_{T(\delta\mathbf{x}_0)} \\ & + \underbrace{\Phi_{(t_1,t)}^{i,k} B_1^k \delta t_1 + \frac{1}{2} \left(\Phi_{(t_1,t)}^{i,k} B_2^k + \Psi_{(t_1,t)}^{i,k_1 k_2} B_1^{k_1} B_1^{k_2} \right) \delta t_1 \delta t_1}_{T(\delta t_1)} + \underbrace{\Phi_{(t_1,t)}^{i,k} \delta\Delta\mathbf{x}_1^k + \frac{1}{2} \Psi_{(t_1,t)}^{i,k_1 k_2} \delta\Delta\mathbf{x}_1^{k_1} \delta\Delta\mathbf{x}_1^{k_2}}_{T(\delta\Delta\mathbf{x}_1)} \\ & + \underbrace{\left(\Phi_{(t_1,t)}^{i,k_1} C^{k_1,p_1} + \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,p_1} B_1^{k_2} \right) \delta x_0^{p_1} \delta t_1}_{T(\delta\mathbf{x}_0, \delta t_1)} + \underbrace{\Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,p_1} I_6^{k_2,p_2} \delta x_0^{p_1} \delta\Delta\mathbf{x}_1^{p_2}}_{T(\delta\mathbf{x}_0, \delta\Delta\mathbf{x}_1)} + \underbrace{\left(\Psi_{(t_1,t)}^{i,k_1 k_2} B_1^{k_1} + \Phi_{(t_1,t)}^{i,k_1} D^{k_1 k_2} \right) \delta\Delta\mathbf{x}_1^{k_2} \delta t_1}_{T(\delta\Delta\mathbf{x}_1, \delta t_1)} \end{aligned} \quad (25)$$

$$\delta\mathbf{x}_1^{+,k} = \delta\mathbf{x}_1^{*,k} + f_+^k \delta t_1 + \frac{1}{2} g_+^k \delta t_1 \delta t_1 + H_+^{k,p} \delta t_1 \delta\mathbf{x}_1^{*,p} = \delta\mathbf{x}_1^{-,k} + \delta\Delta\mathbf{x}_1^k \quad (18)$$

where

$$f_+^k = \frac{\partial x_1^{+,k}}{\partial t_1} = y^k(\mathbf{x}_1^+, t_1) \quad (19)$$

$$g_+^k = \frac{\partial^2 x_1^{+,k}}{\partial t_1 \partial t_1} = \frac{\partial y^k(\mathbf{x}_1^+, t_1)}{\partial t_1} \quad (20)$$

$$\begin{aligned} H_+^{k,p} &= \frac{\partial^2 x_1^{+,k}}{\partial t_1 \partial x_1^{+,p}} = \frac{\partial}{\partial t_1} \left(\frac{\partial x_1^{+,k}}{\partial x_1^{+,p}} \right) = \frac{\partial [\Phi_{(t_1,t_1)}^{k,p}(\mathbf{x}_1^+)]}{\partial t_1} \\ &= \dot{\Phi}_{(t_1,t_1)}^{k,p}(\mathbf{x}_1^+) = \frac{\partial y^k(\mathbf{x}_1^+, t_1)}{\partial x_1^{+,p}} \end{aligned} \quad (21)$$

The deviation of the virtual state $\delta\mathbf{x}_1^*$ can be obtained by applying a reversion of series on Eq. (18), which is given as

$$\begin{aligned} \delta\mathbf{x}_1^{*,k} &= \delta\mathbf{x}_1^{-,k} + \delta\Delta\mathbf{x}_1^k - f_+^k \delta t_1 - \frac{1}{2} g_+^k \delta t_1 \delta t_1 + H_+^{k,p} f_+^p \delta t_1 \delta t_1 \\ &\quad - H_+^{k,p} \delta t_1 (\delta\mathbf{x}_1^{-,p} + \delta\Delta\mathbf{x}_1^p) \end{aligned} \quad (22)$$

Equation (22) is the obtained second-order orbit propagator for the second segment. Then, we connect the propagators in the first segment and the second segment. By substituting Eqs. (13–16) into Eq. (22) and comprising linear and quadratic terms, Eq. (22) can be rewritten as

$$\begin{aligned} \delta\mathbf{x}_1^{*,k} &= A_1^{k,p} \delta x_0^p + B_1^k \delta t_1 + \delta\Delta\mathbf{x}_1^k + \frac{1}{2} A_2^{k,p_1 p_2} \delta x_0^{p_1} \delta x_0^{p_2} \\ &\quad + \frac{1}{2} B_2^k \delta t_1 \delta t_1 + C^{k,p} \delta x_0^p \delta t_1 + D^{k,p} \delta\Delta\mathbf{x}_1^p \delta t_1 \end{aligned} \quad (23)$$

where $A_1^{k,p} = \Phi_{(t_0,t_1)}^{k,p}$, $A_2^{k,p_1 p_2} = \Psi_{(t_0,t_1)}^{k,p_1 p_2}$, $B_1^k = f_-^k - f_+^k$, $B_2^k = g_-^k - g_+^k + 2H_+^{k,p} f_+^p - 2H_+^{k,p} f_-^p$, $C^{k,p} = H_-^{k,p} - H_+^{k,p} \Phi_{(t_0,t_1)}^{k,p}$, and $D^{k,p} = -H_+^{k,p}$.

Finally, we investigate the propagation in the third segment. Similar to Eq. (11), in the period $t_1 \leq t \leq t_2$, the deviation $\delta\mathbf{x}(t)$ can be approximated as

$$\delta\mathbf{x}^i(t) = \Phi_{(t_1,t)}^{i,k} \delta\mathbf{x}_1^{*,k} + \frac{1}{2} \Psi_{(t_1,t)}^{i,k_1 k_2} \delta\mathbf{x}_1^{*,k_1} \delta\mathbf{x}_1^{*,k_2} \quad (24)$$

which, after substituting for $\mathbf{x}_1^*$ from Eq. (23), yields

where I_6 is a six-dimensional identity matrix; $I_6^{k_2,p_2}$ is the element in row k_2 and column p_2 , and $T(\cdot)$ is the independent term of the corresponding variables. One can see from Eqs. (23) and (24) that the virtual state $\mathbf{x}_1^*$ is an important variable in the developed modified STT. Although the virtual state $\mathbf{x}_1^*$ is finally removed from the equations [as shown in Eq. (25)], it helps build a relationship between $\delta\mathbf{x}(t)$ ($t > t_1$) and $\delta\mathbf{x}_0$, $\delta\Delta\mathbf{v}_1$, and δt_1 . The difficulty of developing such a modified STT lies in the coupling effects of $\delta\Delta\mathbf{v}_1$ and δt_1 on orbit deviations. By introducing the virtual state, the proposed modified STT can independently and sequentially address the effects of $\delta\Delta\mathbf{v}_1$ [using Eq. (17)] and δt_1 [using Eqs. (21–23)].

Equation (25) is the obtained second-order solution for orbit propagation. For convenience, let $\delta\mathbf{X} = [\delta\mathbf{x}_0; \delta\Delta\mathbf{v}_1; \delta t_1]$ be the deviation of the estimation parameter vector. Then, rewrite Eqs. (11) and (25) as

$$\delta\mathbf{x}(t) = \Phi(t) \delta\mathbf{X} + \frac{1}{2} \Psi(t) \otimes \delta\mathbf{X} \otimes \delta\mathbf{X} \quad (26)$$

where $\Phi \in \mathbb{R}^{6 \times 10}$ and $\Psi \in \mathbb{R}^{6 \times 10 \times 10}$ denote the proposed modified first-order and second-order STTs, respectively, and the operator $\otimes$ denotes the Kronecker tensor product.

During $[t_0, t_1]$, the modified STTs $\Phi(t)$ and $\Psi(t)$ are given as

$$\Phi^{i,:}(t) = \left[\Phi_{(t_0,t)}^{i,1} \quad \cdots \quad \Phi_{(t_0,t)}^{i,6} \mid \mathbf{0}_{1 \times 3} \mid \mathbf{0}_{1 \times 1} \right] \quad (27)$$

$$\Psi^{i,:,:}(t) = \begin{bmatrix} \Psi_{(t_0,t)}^{i,1,1} & \cdots & \Psi_{(t_0,t)}^{i,1,6} & \vdots & \mathbf{0}_{6 \times 3} & \mathbf{0}_{6 \times 1} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ \Psi_{(t_0,t)}^{i,6,1} & \cdots & \Psi_{(t_0,t)}^{i,6,6} & \vdots & \mathbf{0}_{3 \times 6} & \mathbf{0}_{3 \times 1} \\ \hline \mathbf{0}_{3 \times 6} & \mathbf{0}_{3 \times 3} & \mathbf{0}_{3 \times 1} \\ \hline \mathbf{0}_{1 \times 6} & \mathbf{0}_{1 \times 3} & \mathbf{0}_{1 \times 1} \end{bmatrix} \quad (28)$$

During $[t_1, t_2]$, the modified STTs $\Phi(t)$ and $\Psi(t)$ are partitioned as

$$\Phi^{i,:}(t) = \left[\Phi_{(t_1,t)}^{i,k} A_1^{k,1} \quad \dots \quad \Phi_{(t_1,t)}^{i,k} A_1^{k,6} \mid \Phi_{(t_1,t)}^{i,4} \quad \dots \quad \Phi_{(t_1,t)}^{i,6} \mid \Phi_{(t_1,t)}^{i,k} B_1^k \right] \quad (29)$$

$$\Psi^{i,:,:}(t) = \begin{bmatrix} \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,1} A_1^{k_2,1} & \dots & \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,1} A_1^{k_2,6} & \mid & \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,1} I_6^{k_2,4} & \dots & \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,1} I_6^{k_2,6} & \mid & \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,1} B_1^{k_2} & \mid & \Psi_{(t_1,t)}^{i,k_1} C_{k_1,1} \\ + \Phi_{(t_1,t)}^{i,k} A_2^{k,1,1} & & + \Phi_{(t_1,t)}^{i,k} A_2^{k,1,6} & & & & & & + \Phi_{(t_1,t)}^{i,k_1} C_{k_1,1} \\ \vdots & \ddots & \vdots & & \vdots & \ddots & \vdots & & \vdots \\ \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,6} A_1^{k_2,1} & \dots & \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,6} A_1^{k_2,6} & \mid & \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,6} I_6^{k_2,4} & \dots & \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,6} I_6^{k_2,6} & \mid & \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,6} B_1^{k_2} & \mid & \Psi_{(t_1,t)}^{i,k_1} C_{k_1,6} \\ + \Phi_{(t_1,t)}^{i,k} A_2^{k,6,1} & & + \Phi_{(t_1,t)}^{i,k} A_2^{k,6,6} & & & & & & + \Phi_{(t_1,t)}^{i,k_1} C_{k_1,6} \\ \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,1} I_6^{k_2,4} & \dots & \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,1} I_6^{k_2,6} & \mid & \Psi_{(t_1,t)}^{i,4,4} & \dots & \Psi_{(t_1,t)}^{i,4,6} & \mid & \Psi_{(t_1,t)}^{i,k_1,4} B_1^{k_1} + \Phi_{(t_1,t)}^{i,k_1} D_{k_1,4} \\ \vdots & \ddots & \vdots & & \vdots & \ddots & \vdots & & \vdots \\ \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,1} I_6^{k_2,6} & \dots & \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,1} I_6^{k_2,6} & \mid & \Psi_{(t_1,t)}^{i,6,4} & \dots & \Psi_{(t_1,t)}^{i,6,6} & \mid & \Psi_{(t_1,t)}^{i,k_1,6} B_1^{k_1} + \Phi_{(t_1,t)}^{i,k_1} D_{k_1,6} \\ \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,6} B_1^{k_2} & \dots & \Psi_{(t_1,t)}^{i,k_1 k_2} A_1^{k_1,6} B_1^{k_2} & \mid & \Psi_{(t_1,t)}^{i,k_1,4} B_1^{k_1} & \dots & \Psi_{(t_1,t)}^{i,k_1,6} B_1^{k_1} & \mid & \Psi_{(t_1,t)}^{i,k_1 k_2} B_1^{k_1} B_1^{k_2} \\ + \Phi_{(t_1,t)}^{i,k_1} C_{k_1,1} & & + \Phi_{(t_1,t)}^{i,k_1} C_{k_1,6} & & + \Phi_{(t_1,t)}^{i,k_1} D_{k_1,4} & & + \Phi_{(t_1,t)}^{i,k_1} D_{k_1,6} & & + \Phi_{(t_1,t)}^{i,k} B_2^k \end{bmatrix} \quad (30)$$

A summary of processes in Eqs. (11–30) is provided in Algorithm 1. The pseudocode presented herein is used to clearly describe the derivation process of Eqs. (11–30). It does not mean that the modified STTs are repeatedly computed when predicting the orbit state deviation $\delta\mathbf{x}(t)$. In practice, once the modified STTs are obtained, the orbit state deviation $\delta\mathbf{x}(t)$ can be analytically predicted without additional numerical integrations. It should also be noted that in Eq. (23) the deviation of the virtual state $\delta\mathbf{x}_1^*$, rather than the deviation of the true state $\delta\mathbf{x}(t_1)$, is predicted. The residual between $\delta\mathbf{x}_1^*$ and $\delta\mathbf{x}(t_1)$, in other words, the difference between the red line and green dot in Fig. 1, is neglected by the proposed modified STT [Eqs. (26–30)]. Therefore, the accuracy of the proposed modified STT slightly degrades around the epoch t_1 . The accuracy performance of the proposed modified STT around the epoch t_1 will be discussed in Sec. IV.A.

Algorithm 1: Pseudocode of calculating the modified STT

Inputs: initial epoch t_0 ; final epoch t_2 ; reference variables $\mathbf{x}_0$, t_1 , and $\Delta\mathbf{v}_1$; and the deviations $\delta\mathbf{x}_0$, δt_1 , and $\delta\Delta\mathbf{v}_1$

Output: the orbit state deviation $\delta\mathbf{x}(t)$ at any given epoch t ($t \in [t_0, t_1]$) and the modified STTs $\Phi(t)$ and $\Psi(t)$

- 1 Integrate the STTs $\Phi_{(t_0,t)}^{k,p}$ and $\Psi_{(t_0,t)}^{k,p_1 p_2}$ using Eqs. (8–10).
- 2 Predict $\delta\mathbf{x}(t)$ ($t \in [t_0, t_1]$) and $\delta\mathbf{x}(t_1)$ using Eqs. (11) and (12).
- 3 Calculate $\delta\mathbf{x}_1^-$ using Eqs. (13–16).
- 4 Calculate $\delta\mathbf{x}_1^+$ using Eqs. (18–21).
- 5 Calculate the virtual state deviation $\delta\mathbf{x}_1^*$ using Eq. (23).
- 6 Predict $\delta\mathbf{x}(t)$ ($t \in [t_1, t_2]$) using Eq. (25).
- 7 Calculate the modified STTs $\Phi(t)$ and $\Psi(t)$ using Eqs. (27–30).

B. Second-Order Measurement Model

Equations (26–30) present the relationship between the deviations of the estimation parameter vector $\delta\mathbf{X}$ and $\delta\mathbf{x}(t)$. In this subsection, the transformation from the deviation of the estimation parameter vector $\delta\mathbf{X}$ to the deviation of the measurement $\delta\mathbf{z}(t)$ at any future epoch t is derived. According to Eq. (2), the measurement deviation $\delta\mathbf{z}(t)$ can be approximated using a second-order Taylor series as

$$\delta\mathbf{z}(t) = \mathbf{U}(t)\delta\mathbf{x}(t) + \frac{1}{2}\mathbf{Q}(t) \otimes \delta\mathbf{x}(t) \otimes \delta\mathbf{x}(t) \quad (31)$$

where

$$\mathbf{U}^{i,k}(t) = \frac{\partial h^i(\mathbf{x}(t))}{\partial \mathbf{x}^k(t)} \quad (32)$$

$$\mathbf{Q}^{i,k_1 k_2}(t) = \frac{\partial^2 h^i(\mathbf{x}(t))}{\partial \mathbf{x}^{k_1}(t) \partial \mathbf{x}^{k_2}(t)} \quad (33)$$

Substituting Eq. (26) into Eq. (31) and neglecting higher-order terms,

$$\delta\mathbf{z}(t) = \mathbf{U}(t)\Phi(t)\delta\mathbf{X} + \frac{1}{2}\mathbf{U}(t)[\Psi(t) \otimes \delta\mathbf{X} \otimes \delta\mathbf{X}] + \frac{1}{2}\mathbf{Q}(t) \otimes (\Phi(t)\delta\mathbf{X}) \otimes (\Phi(t)\delta\mathbf{X}) \quad (34)$$

Equation (34) can be further simplified as

$$\delta\mathbf{z}(t) = \Xi(t)\delta\mathbf{X} + \frac{1}{2}\Theta(t) \otimes \delta\mathbf{X} \otimes \delta\mathbf{X} \quad (35)$$

where

$$\Xi^{i,p}(t) = \mathbf{U}^{i,k}(t)\Phi^{k,p}(t) \quad (36)$$

$$\Theta^{i,p_1 p_2}(t) = \mathbf{U}^{i,j}(t)\Psi^{j,p_1 p_2}(t) + \mathbf{Q}^{i,k_1 k_2}(t)\Phi^{k_1,p_1}(t)\Phi^{k_2,p_2}(t) \quad (37)$$

Equations (35–37) are the derived second-order models that transform from the deviation of the estimation parameter vector $\delta\mathbf{X}$ to the measurement deviation $\delta\mathbf{z}(t)$. Assume that the estimation parameter vector deviation $\delta\mathbf{X}$ is a zero-mean random vector, and its covariance matrix is represented by $\mathbf{P} = [P^{ab}]_{10 \times 10}$. The mean $\mathbf{m}(\delta\mathbf{z}(t)) = [\mathbf{m}^i(\delta\mathbf{z}(t))]_{d \times 1}$ and covariance matrix $\mathbf{P}(\delta\mathbf{z}(t)) = [P^{ij}(\delta\mathbf{z}(t))]_{d \times d}$ of the measurement deviation $\delta\mathbf{z}(t)$ are then given as

$$\mathbf{m}^i(\delta\mathbf{z}(t)) = \frac{1}{2} \sum_{a=1}^n \sum_{b=1}^n \Theta^{i,ab}(t) P^{ab} \quad (38)$$

$$\begin{aligned} P^{ij}(\delta\mathbf{z}(t)) &= \sum_{a=1}^{10} \sum_{\alpha=1}^{10} \Xi^{i,a}(t) \Xi^{j,\alpha}(t) P^{a\alpha} - \mathbf{m}^i(\delta\mathbf{z}(t)) \mathbf{m}^j(\delta\mathbf{z}(t)) \\ &+ \frac{1}{4} \sum_{a=1}^{10} \sum_{b=1}^{10} \sum_{\alpha=1}^{10} \sum_{\beta=1}^{10} \Theta^{i,ab}(t) \Theta^{j,\alpha\beta}(t) [P^{ab} P^{\alpha\beta} \\ &+ P^{a\alpha} P^{b\beta} + P^{a\beta} P^{b\alpha}] \end{aligned} \quad (39)$$

For the sake of simplicity, the nonlinear covariance propagations in Eqs. (38) and (39) are denoted as

$$[\mathbf{m}(\delta\mathbf{z}(t)), \mathbf{P}(\delta\mathbf{z}(t))] = \text{propagation}(\mathbf{P}, \Xi(t), \Theta(t)) \quad (40)$$

C. Second-Order Optimal Estimation

Starting from a priori estimations (obtained from initial, or prior, orbit determination), the reference orbit can be propagated using the modified STT in Eq. (26), and the residuals between the incoming observation data and the predicted measurements are obtained. Based on the partial derivatives of the measurements with respect to the estimation parameter vector $\mathbf{X}$ in Eqs. (35–37), a priori estimations can be corrected.

Assume that, in the period $[t_0, t_2]$, K observations are obtained using the equipped sensor. The K observations are labeled by

$$\tilde{\mathbf{z}}(\tau_j) = \mathbf{h}(\tilde{\mathbf{x}}(\tau_j)) + \boldsymbol{\varepsilon}, j \in \{1, 2, \dots, K\} \quad (41)$$

where $t_0 \leq \tau_1 < \tau_2 < \dots < \tau_{K-1} < \tau_K \leq t_2$ are the epoch series when the observations are received. Note that the reference orbit state $\mathbf{x}(t)$ can be computed based on the current estimation $\mathbf{X} = [\mathbf{x}_0; \Delta \mathbf{v}_1; t_1]$. The predicted measurement $\mathbf{z}(\tau_i) = \mathbf{h}(\mathbf{x}(\tau_i))$ is then calculated using the reference orbit state $\mathbf{x}(\tau_i)$ and the measurement model in Eq. (2). Thus, the residuals of the K observations are written as

$$\begin{aligned} \delta \mathbf{Z} &= \begin{bmatrix} \delta \mathbf{z}(\tau_1) \\ \delta \mathbf{z}(\tau_2) \\ \vdots \\ \delta \mathbf{z}(\tau_K) \end{bmatrix} = \begin{bmatrix} \tilde{\mathbf{z}}(\tau_1) - \mathbf{z}(\tau_1) \\ \tilde{\mathbf{z}}(\tau_2) - \mathbf{z}(\tau_2) \\ \vdots \\ \tilde{\mathbf{z}}(\tau_K) - \mathbf{z}(\tau_K) \end{bmatrix} = \delta \mathbf{h} + \boldsymbol{\varepsilon} \\ &= \begin{bmatrix} \mathbf{h}(\tilde{\mathbf{x}}(\tau_1)) - \mathbf{h}(\mathbf{x}(\tau_1)) \\ \mathbf{h}(\tilde{\mathbf{x}}(\tau_2)) - \mathbf{h}(\mathbf{x}(\tau_2)) \\ \vdots \\ \mathbf{h}(\tilde{\mathbf{x}}(\tau_K)) - \mathbf{h}(\mathbf{x}(\tau_K)) \end{bmatrix} + \begin{bmatrix} \boldsymbol{\varepsilon}_1 \\ \boldsymbol{\varepsilon}_2 \\ \vdots \\ \boldsymbol{\varepsilon}_K \end{bmatrix} \end{aligned} \quad (42)$$

where $\boldsymbol{\varepsilon}_i \in \mathbb{R}^d$ is the measurement noise of the i th observation.

Let $\delta \mathbf{X} \in \mathbb{R}^{10}$ be the deviation between the real state $\tilde{\mathbf{X}} = [\tilde{\mathbf{x}}_0; \Delta \tilde{\mathbf{v}}_1; \tilde{t}_1] \in \mathbb{R}^{10}$ and the current estimation $\mathbf{X}$. According to Eqs. (31–37), $\delta \mathbf{h}$ can be approximated using a series of second-order expressions as

$$\delta \mathbf{h} = \delta \mathbf{Z}' = \begin{bmatrix} \delta \mathbf{z}'(\tau_1) \\ \delta \mathbf{z}'(\tau_2) \\ \vdots \\ \delta \mathbf{z}'(\tau_K) \end{bmatrix} = \begin{bmatrix} \Xi(\tau_1)\delta \mathbf{X} + \frac{1}{2}\Theta(\tau_1) \otimes \delta \mathbf{X} \otimes \delta \mathbf{X} \\ \Xi(\tau_2)\delta \mathbf{X} + \frac{1}{2}\Theta(\tau_2) \otimes \delta \mathbf{X} \otimes \delta \mathbf{X} \\ \vdots \\ \Xi(\tau_K)\delta \mathbf{X} + \frac{1}{2}\Theta(\tau_K) \otimes \delta \mathbf{X} \otimes \delta \mathbf{X} \end{bmatrix} \quad (43)$$

The expression of $\delta \mathbf{Z}'$ in Eq. (43) can be further simplified as

$$\delta \mathbf{Z}' = \Omega \delta \mathbf{X} + \frac{1}{2} \Sigma \otimes \delta \mathbf{X} \otimes \delta \mathbf{X} \quad (44)$$

where

$$\Omega^{i,p} = [\Xi^{i,p}(\tau_1); \Xi^{i,p}(\tau_2); \dots; \Xi^{i,p}(\tau_K)] \quad (45)$$

$$\Sigma^{i,p_1 p_2} = [\Theta^{i,p_1 p_2}(\tau_1); \Theta^{i,p_1 p_2}(\tau_2); \dots; \Theta^{i,p_1 p_2}(\tau_K)] \quad (46)$$

In this paper, the goal of the OD of the spacecraft under an impulsive maneuver is to determine the optimal $\delta \mathbf{X}$ that can minimize the deviation between $\delta \mathbf{Z}$ and $\delta \mathbf{Z}'$. In the remainder of this subsection, the second-order optimal estimation, extended from the conventional least-squares solution, is presented to solve the impulsive OD method.

Substituting Eqs. (43) and (44) into Eq. (42) yields

$$\delta \mathbf{Z} = \delta \mathbf{h} + \boldsymbol{\varepsilon} = \Omega \delta \mathbf{X} + \frac{1}{2} \Sigma \otimes \delta \mathbf{X} \otimes \delta \mathbf{X} + \boldsymbol{\varepsilon} \quad (47)$$

First, the weights of each term should be determined. Note that the covariance matrix of $\delta \mathbf{z}'(\tau_i)$ is

$$\mathbf{P}_i' = \text{Propagation}(\mathbf{P}(\delta \mathbf{X}), \Xi(\tau_i), \Theta(\tau_i)) \quad (48)$$

which can be obtained using Eqs. (38) and (39). Thus, the uncertainty of $\delta \mathbf{z}(\tau_i) - \delta \mathbf{z}'(\tau_i)$ can be represented by a covariance matrix $\mathbf{P}_i$ as

$$\mathbf{P}_i = \mathbf{P}_i' + \mathbf{R} \quad (49)$$

Thus, the weights matrix $\mathbf{W}$ is represented by

$$\mathbf{W} = \begin{bmatrix} \mathbf{P}_1 & \mathbf{0}_{d \times (K-2)d} & \mathbf{0}_{d \times d} \\ \mathbf{0}_{(K-2)d \times d} & \ddots & \mathbf{0}_{(K-2)d \times d} \\ \mathbf{0}_{d \times d} & \mathbf{0}_{d \times (K-2)d} & \mathbf{P}_K \end{bmatrix}^{-1} \quad (50)$$

To avoid a numerical singularity when obtaining the optimal solution, $\mathbf{W}$ can be normalized as $\mathbf{W}/\|\mathbf{W}\|$ when solving the OD problem. Combining Eqs. (42–44) with Eq. (50), the performance metric to be minimized is defined as

$$\begin{aligned} J &= \left(\delta \mathbf{Z} - \Omega \delta \mathbf{X} - \frac{1}{2} \Sigma \otimes \delta \mathbf{X} \otimes \delta \mathbf{X} \right)^T \\ &\quad \times \mathbf{W} \left(\delta \mathbf{Z} - \Omega \delta \mathbf{X} - \frac{1}{2} \Sigma \otimes \delta \mathbf{X} \otimes \delta \mathbf{X} \right) \\ &= \text{Tr} \left(\left(\delta \mathbf{Z} - \Omega \delta \mathbf{X} - \frac{1}{2} \Sigma \otimes \delta \mathbf{X} \otimes \delta \mathbf{X} \right) \mathbf{W}^T \right. \\ &\quad \left. \times \left(\delta \mathbf{Z} - \Omega \delta \mathbf{X} - \frac{1}{2} \Sigma \otimes \delta \mathbf{X} \otimes \delta \mathbf{X} \right)^T \right) \end{aligned} \quad (51)$$

Similar to the deviation of the LSM, the partial derivative of the metrics J can be obtained by

$$\frac{\partial J}{\partial \mathbf{X}} = -2\mathbf{\Gamma}^T \mathbf{W} \left(\delta \mathbf{Z} - \Omega \delta \mathbf{X} - \frac{1}{2} \Sigma \otimes \delta \mathbf{X} \otimes \delta \mathbf{X} \right) \quad (52)$$

where

$$\mathbf{\Gamma}^{i,k} = \Omega^{i,k} + \Sigma^{i,kp} \delta \mathbf{X}^p \quad (53)$$

The optimal estimation $\delta \mathbf{X}$ can be obtained by solving the equation $\partial J / \partial \mathbf{X} = \mathbf{0}_{1 \times 10}$. However, the solution cannot be directly computed as Eq. (52) is a high-order system. A method to obtain the second-order optimal estimation of $\delta \mathbf{X}$ is therefore proposed herein.

Neglecting the second-order terms of Eq. (44), the performance metric J is rewritten as

$$\bar{J} = (\delta \mathbf{Z} - \Omega \delta \mathbf{X})^T \mathbf{W} (\delta \mathbf{Z} - \Omega \delta \mathbf{X}) = \text{Tr}((\delta \mathbf{Z} - \Omega \delta \mathbf{X}) \mathbf{W}^T (\delta \mathbf{Z} - \Omega \delta \mathbf{X})^T) \quad (54)$$

where $\bar{J}$ is the performance metric containing only first-order terms.

By solving the equation $\partial \bar{J} / \partial \mathbf{X} = \mathbf{0}_{1 \times 10}$, a first-order (linear) optimal solution is obtained by

$$\delta \bar{\mathbf{X}} = (\Omega^T \mathbf{W} \Omega)^{-1} \Omega^T \mathbf{W} \delta \mathbf{Z} \quad (55)$$

Substituting the first-order solution into Eq. (53), the first-order approximation of $\mathbf{\Gamma}$ is calculated as

$$\bar{\mathbf{\Gamma}}^{i,k} = \Omega^{i,k} + \Sigma^{i,kp} \delta \bar{\mathbf{X}}^p \quad (56)$$

Substituting the approximation in Eq. (56) into the partial derivative of the metrics J , Eq. (52) is written as

$$\frac{\partial J}{\partial \bar{X}} = -2\bar{\Gamma}^T W \left(\delta Z - \Omega \delta X - \frac{1}{2} \Sigma \otimes \delta X \otimes \delta X \right) = \mathbf{0}_{1 \times 10} \quad (57)$$

Similar to the operations in Eqs. (34–37), Eq. (57) can be simplified as

$$\delta \bar{Z} - \bar{\Omega} \delta X - \frac{1}{2} \bar{\Sigma} \otimes \delta X \otimes \delta X = \mathbf{0}_{1 \times 10} \quad (58)$$

where

$$\delta \bar{Z}^i = -2\bar{\Gamma}^{ji} W^{jk} \delta Z^k \quad (59)$$

$$\bar{\Omega}^{i,p} = -2\bar{\Gamma}^{ji} W^{jk} \Omega^{kp} \quad (60)$$

$$\bar{\Sigma}^{i,p_1 p_2} = -2\bar{\Gamma}^{ji} W^{jk} \Sigma^{k,p_1 p_2} \quad (61)$$

According to Turner's method of series reversion [45], the solution of Eq. (58) is approximated by

$$\delta \hat{X} = \bar{\Omega}^{-1} \delta \bar{Z} - \frac{1}{2} \bar{\Omega}^{-1} [\bar{\Sigma} \otimes (\bar{\Omega}^{-1} \delta \bar{Z}) \otimes (\bar{\Omega}^{-1} \delta \bar{Z})] \quad (62)$$

where $\delta \hat{X}$ is the approximated second-order optimal estimation. Equation (62) is the proposed second-order optimal estimation of the parameter of interest. Using Eq. (62), the estimation X can be corrected as $X \leftarrow X + \delta \hat{X}$.

Comprising only linear and quadratic terms, the associated covariance matrix of the estimation $\delta \hat{X}$ is given as

$$P(\delta \hat{X}) = E[(\delta X - \delta \hat{X})(\delta X - \delta \hat{X})^T] \quad (63)$$

Substitution of Eq. (62) into Eq. (63) results in

$$\delta X - \delta \hat{X} = \delta X - \bar{\Omega}^{-1} \delta \bar{Z} + \frac{1}{2} \bar{\Omega}^{-1} [\bar{\Sigma} \otimes (\bar{\Omega}^{-1} \delta \bar{Z}) \otimes (\bar{\Omega}^{-1} \delta \bar{Z})] \quad (64)$$

Substitute Eq. (47) into Eq. (64), and reserve only the first- and second-order terms. Then, $\delta X - \delta \hat{X}$ can be expressed as

$$\begin{aligned} \delta X - \delta \hat{X} &= \delta X - (\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W \delta Z \\ &\quad + \frac{1}{2} (\bar{\Gamma}^T W \Omega)^{-1} \{ \bar{\Gamma}^T W \Sigma \otimes [(\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W \delta Z] \\ &\quad \otimes [(\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W \delta Z] \} \\ &= \delta X - (\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W \left(\Omega \delta X + \frac{1}{2} \Sigma \otimes \delta X \otimes \delta X + \epsilon \right) \\ &\quad + \frac{1}{2} (\bar{\Gamma}^T W \Omega)^{-1} \{ \bar{\Gamma}^T W \Sigma \otimes [\delta X + (\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W \epsilon] \\ &\quad \otimes [\delta X + (\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W \epsilon] \} \\ &= -(\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W \epsilon + \frac{1}{2} (\bar{\Gamma}^T W \Omega)^{-1} \{ \bar{\Gamma}^T W \Sigma \otimes \delta X \\ &\quad \otimes [(\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W \epsilon] \} \\ &\quad + \frac{1}{2} (\bar{\Gamma}^T W \Omega)^{-1} \{ \bar{\Gamma}^T W \Sigma \otimes [(\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W \epsilon] \otimes \delta X \} \\ &\quad + \frac{1}{2} (\bar{\Gamma}^T W \Omega)^{-1} \{ \bar{\Gamma}^T W \Sigma \otimes [(\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W \epsilon] \\ &\quad \otimes [(\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W \epsilon] \} \end{aligned} \quad (65)$$

Substituting the results in Eq. (65) into Eq. (63), the covariance matrix of the second-order estimation $\delta \hat{X}$ is further given as

$$\begin{aligned} P(\delta \hat{X}) &= \mathbb{E} \{ [(\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W \epsilon] [(\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W \epsilon]^T \} \\ &= \mathbb{E} [(\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W \epsilon \epsilon^T W^T \bar{\Gamma}^T (\Omega^T W \bar{\Gamma})^{-1}] \\ &= (\bar{\Gamma}^T W \Omega)^{-1} \bar{\Gamma}^T W R W^T \bar{\Gamma} (\Omega^T W \bar{\Gamma})^{-1} \end{aligned} \quad (66)$$

Note that the second-order solution in Eq. (62) corresponds to one round of iteration. The estimations in one iteration are not sufficiently accurate. Thus, to achieve a certain threshold level, it is necessary to carry out multiple iterations so that the algorithm can converge. The convergence performance of the proposed method will be presented in the simulation section.

D. Overall Procedure

A flow diagram outlining the overall procedure of the proposed method for orbit estimation and maneuver reconstruction is illustrated in Fig. 2. The inputs of the algorithm include the initial

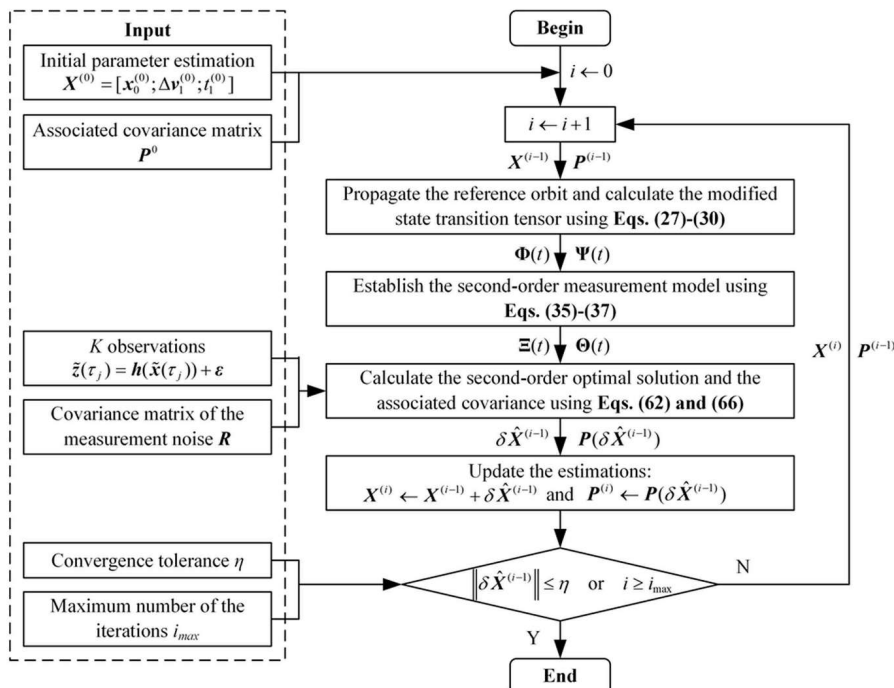


Fig. 2 A flow diagram of the proposed method.

parameter estimation $\mathbf{X}^{(0)} = [\mathbf{x}_0^{(0)}; \Delta \mathbf{v}_1^{(0)}; t_1^{(0)}]$ and its associated covariance matrix $\mathbf{P}^0$, the K observations $\tilde{\mathbf{z}}(\tau_j) j \in \{1, 2, \dots, K\}$, and the covariance matrix of the measurement noise $\mathbf{R}$. The superscript (i) denotes the results of the i th iteration. It differs from the previous superscripts that represent the elements [for example, the superscripts in Eqs. (59–61)] in that it has parentheses. The iterations terminate if the Frobenius norm of the optimal solution $\delta \hat{\mathbf{X}}^{(i-1)}$ is smaller than the predefined tolerance η or the maximum iteration $i_{\max}$ is reached.

Note that the proposed method is based on the second-order modified STT [Eqs. (26–30)] and the second-order optimal solution [Eqs. (62) and (66)]. The linear version of the solution can be found in Eq. (55). One of the advantages of such a second-order model is that the coupling term of the epoch t_1 and the vector $\Delta \mathbf{v}_1$ is considered [the term $\mathbf{T}(\delta \Delta \mathbf{x}_1, \delta t_1)$ in Eq. (25)]. The epoch and vector of the impulsive maneuver are critical to the impulsive maneuvering spacecraft OD problem; thus, considering the coupling term can improve accuracy and convergence of the algorithm. A comparison between the linear and second-order solutions will be given in Sec. IV.A.1.

IV. Numerical Simulations

To demonstrate protentional use cases for the equations derived in Sec. III, two simple applications, one in the low-Earth orbit (LEO) and the other in the geostationary orbit (GEO), are first studied. The orbit estimation and maneuver reconstruction performances of the proposed method are analyzed by comparing with several conventional maneuvering spacecraft OD algorithms. Then, sensitivity analyses are performed to investigate the influences of the initial guess, magnitude of the maneuver, and measurement interval on method performances.

A. Estimated Accuracy Analysis

1. LEO Testing Case

Consider two spacecraft in low-Earth orbit, with one being the target and the other being the observer. The nominal orbit elements of the target spacecraft and observer spacecraft are listed in Table 1, where h , e , n , i , ω , and Ω label the altitude at apogee, eccentricity, true anomaly, inclination, argument of periapsis, and longitude of the ascending node, respectively.

Let $\mathbf{x}_O = [\mathbf{r}_O; \mathbf{v}_O] = [x_O, y_O, z_O, \dot{x}_O, \dot{y}_O, \dot{z}_O]^T$ denote the orbit state of the observer spacecraft. Assume the observer observes the target using onboard optical sensors. The line-of-sight (LOS) vector from the observer to the target can be obtained by identifying points

in the image and extracting the target [46,47]. In this paper, the adopted model for representing the angular measurements is given as

$$\mathbf{l} = [l_x, l_y, l_z]^T = \frac{\mathbf{r} - \mathbf{r}_O}{\|\mathbf{r} - \mathbf{r}_O\|} + \boldsymbol{\varepsilon}_l \in \mathbb{R}^3 \quad (67)$$

where $\boldsymbol{\varepsilon}_l \in \mathbb{R}^3$ is the measurement noise vector and $\mathbf{l} = [l_x, l_y, l_z]^T$ denotes the LOS vector, with l_x , l_y , and l_z representing the elements along the inertial x , y , and z axes, respectively.

Three cases are simulated: one standard case, one sparse observation case, and one short-arc case. The conditions of these three cases are listed in Table 2. The total navigation period of the standard case is 1800 s, with a measurement interval of 10 s. The standard case is used to test the performance of the proposed method in a general condition. In addition, the robustness of the proposed method under severe conditions is investigated using the sparse observation case and the short-arc observation case. In the sparse observation case, only 11 measurements are available, which is used to analyze whether the method is effective for conditions when information is limited. The third case has a total navigation period of 300 s, in which the performance of the method under a short observation arc is tested.

Before testing the performance of the proposed method, the accuracy of the proposed modified STT in Eq. (26) and the second-order measurement model in Eq. (35) is analyzed. The standard case is employed for accuracy analysis. The process of testing the accuracy of the modified STT and the second-order measurement model is based on 1000 Monte Carlo (MC) runs. Each sample $\tilde{\mathbf{X}}$ is generated randomly from the normal distribution $\mathcal{N}(\tilde{\mathbf{X}}; \mathbf{X}_r, \mathbf{P}_X)$, with $\mathbf{X}_r$ being the real state listed in Table 1 and $\mathbf{P}_X$ being the covariance

$$\mathbf{P}_X = \text{diag}([100^2 \text{ km}^2, 100^2 \text{ km}^2, 100^2 \text{ km}^2, 10^2 \text{ m}^2/\text{s}^2, 10^2 \text{ m}^2/\text{s}^2, 10^2 \text{ m}^2/\text{s}^2, 5^2 \text{ m}^2/\text{s}^2, 5^2 \text{ m}^2/\text{s}^2, 5^2 \text{ m}^2/\text{s}^2, 50^2 \text{ s}^2]) \quad (68)$$

For each sample $\tilde{\mathbf{X}}$, the true orbit state $\tilde{\mathbf{x}}(t)$ from t_0 to t_2 is propagated, and the state deviation $\delta \mathbf{x}(t)$ is obtained. Moreover, the measurement deviation $\delta \mathbf{z}(t)$ is calculated. The state deviation $\delta \mathbf{x}(t)$ and the measurement deviation $\delta \mathbf{z}(t)$ are also predicted using Eqs. (26) and (35), and the absolute errors (AEs) and the relative errors are obtained. The time history of the mean relative errors (MREs) of the 1000 MC runs is shown in Fig. 3. In Fig. 3, the green lines represent the impulsive maneuver epoch. The MREs of the position states are smaller than 0.35%, and the MREs of the velocity states are no larger than 6%. The position and velocity vectors have spikes in MRE around the maneuver epoch (i.e., the green lines). Remember that in Eq. (23) the residual between $\delta \mathbf{x}_1^*$ and $\delta \mathbf{x}(t_1)$ is neglected by our equations, making the accuracy of the proposed modified STT slightly degrade around the epoch t_1 . However, a MRE of 6% is applicable for the modified STT to be used in orbit estimation and maneuver reconstruction.

To better display the predicted errors of the second-order measurement model, the LOS vectors are converted to angles $\alpha(\mathbf{l})$ and $\beta(\mathbf{l})$, defined as

Table 1 Nominal orbit elements of the target and the observer in the LEO testing case

Spacecraft	h , km	e	i , deg	Ω , deg	ω , deg	n , deg
Observer	500	0.01	45.05	29.93	132.9	−107.74
Target	1000	0.02	45.00	94.80	199.00	−54.13

Table 2 Simulated conditions of three cases

Parameter	Standard case	Sparse observation case	Short-arc observation case
Position	10 ² km ² per axis		
Velocity	1 ² m ² /s ² per axis		
Impulsive vector	5 ² m ² /s ² per axis		
Impulsive epoch	50 ² s ²	10 ² s ²	
Standard deviation of measurement noise	1 × 10 ^{−5} per axis		
Measurement frequency	10 s	180 s	2 s
Start epoch t_0	0 s		
End epoch t_2	1800 s	1800 s	300 s
Impulsive maneuver epoch t_1	905 s	905 s	151 s
Impulsive maneuver vector $\Delta \mathbf{v}_1$		[10, 10, 10] ^T m/s	

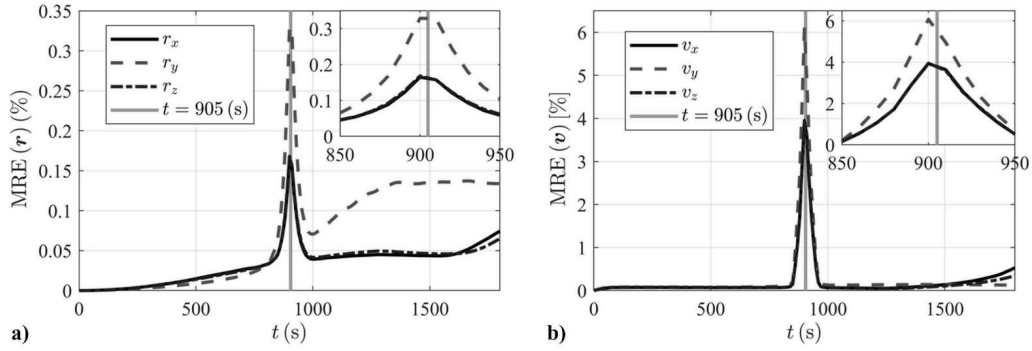


Fig. 3 Time history of the MRE of the modified STT: a) position and b) velocity.

$$\alpha(I) = \sin^{-1}(l_z/\|I\|) = \sin^{-1} l_z \quad (69)$$

$$\beta(I) = \tan^{-1}(l_y/l_x) \quad (70)$$

The accuracy testing results of the second-order measurement model are shown in Fig. 4. Figures 4a and 4b illustrate the time history of the mean absolute error (MAE) and MRE results of 1000 MC runs. The MREs of $\alpha(I)$ and $\beta(I)$ are smaller than 0.2%.

An independent simulation is first performed to show the convergence process of the proposed method. In the independent simulation, the position and velocity estimation errors of the initial guess are set to 10 km per axis and 1 m/s per axis, respectively. The impulsive vector estimation errors are set to be 5 m/s per axis. The

impulsive epoch estimation error is given as 50 s. The standard deviation (STD) of the LOS measurements is set to be 10^{-5} per axis (approximately 2 arcs). The tolerance is set as $\eta = 10^{-3}$, and the maximal iteration is set to be $i_{\max} = 10$. The algorithm converges after five iterations. Table 3 shows the estimated errors of each iteration. The position and velocity estimated errors are smaller than 0.23 km and 0.34 m/s, respectively. In addition, the proposed approach can provide estimations of the unknown impulsive magnitude and epoch with errors no larger than 0.3 m/s and 0.35 s, respectively.

The same independent simulation is carried out using the linear solution in Eq. (55). The linear algorithm converges using eight iterations, with the results detailed in Table 4. The position and

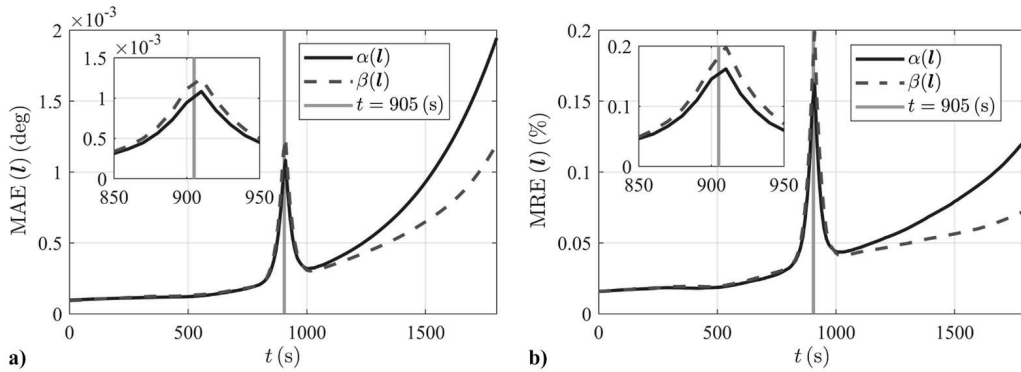


Fig. 4 Time history of the MRE of the second-order measurement model: a) MAE and b) MRE.

Table 3 Estimation errors of the second-order solution in one simulation of the standard case

	Position, km			Velocity, m/s			Impulsive vector, m/s			Epoch, s
	$x(t_0)$	$y(t_0)$	$z(t_0)$	$\dot{x}(t_0)$	$\dot{y}(t_0)$	$\dot{z}(t_0)$	Δv_x	Δv_y	Δv_z	
Initial guess	10	10	10	1	1	1	5	5	5	50
Iteration 1	-0.4714	-0.2973	0.0704	0.7431	-0.3589	-0.4346	-0.6602	0.8008	0.0316	8.2821
Iteration 2	-0.1981	-0.1066	0.0129	0.2929	-0.2040	-0.1040	0.0201	0.3238	-0.1968	-0.3289
Iteration 3	-0.2242	-0.1313	0.0128	0.3294	-0.1843	-0.1082	0.0228	0.2927	-0.2124	0.3388
Iteration 4	-0.2249	-0.1318	0.0129	0.3305	-0.1844	-0.1085	0.0226	0.2929	-0.2128	0.3455
Iteration 5	-0.2248	-0.1318	0.0129	0.3304	-0.1844	-0.1085	0.0226	0.2929	-0.2128	0.3457

Table 4 Estimation errors of the linear solution in one simulation of the standard case

	Position, km			Velocity, m/s			Impulsive vector, m/s			Epoch, s
	$x(t_0)$	$y(t_0)$	$z(t_0)$	$\dot{x}(t_0)$	$\dot{y}(t_0)$	$\dot{z}(t_0)$	Δv_x	Δv_y	Δv_z	
Initial guess	10	10	10	1	1	1	5	5	5	50
Iteration 1	-5.8014	-4.0957	0.9770	8.9495	-3.2535	-3.7784	2.9264	3.6605	-2.8009	40.3799
Iteration 2	-1.1504	-0.8357	0.1950	1.7704	-0.4382	-0.6944	-0.1674	0.4406	-0.4714	8.7904
Iteration 3	-0.4955	-0.3501	0.0906	0.7761	-0.1350	-0.3091	-0.0425	0.2629	-0.3175	3.5572
Iteration 4	-0.4396	-0.3136	0.0794	0.6895	-0.0872	-0.2701	-0.0622	0.1965	-0.2814	3.4616
Iteration 5	-0.3156	-0.2258	0.0557	0.4969	-0.0155	-0.1859	-0.1219	0.2274	-0.2155	2.8066
Iteration 6	-0.3038	-0.2175	0.0534	0.4790	-0.0081	-0.1778	-0.1284	0.2219	-0.2088	2.7618
Iteration 7	-0.3033	-0.2171	0.0533	0.4782	-0.0078	-0.1774	-0.1287	0.2217	-0.2085	2.7604
Iteration 8	-0.3033	-0.2171	0.0533	0.4782	-0.0078	-0.1774	-0.1287	0.2216	-0.2085	2.7603

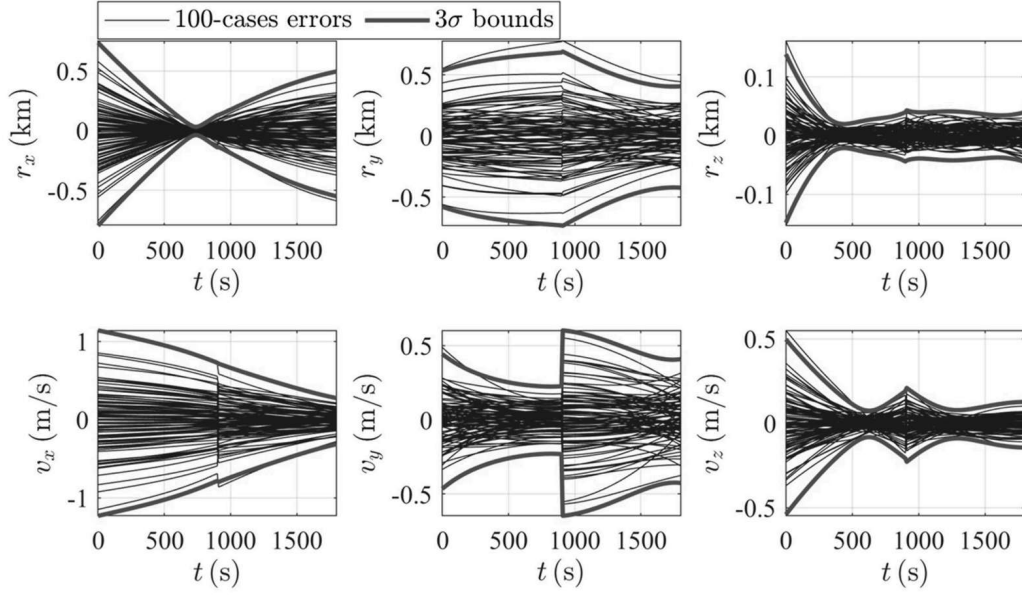


Fig. 5 Monte Carlo simulation results of the proposed method in the standard case.

velocity estimated errors are more than 0.3 km and 0.47 m/s, which are 1.5 times larger than those in Table 3. Additionally, the impulsive epoch errors of the linear solution (2.7603 s) are approximately one time of magnitude larger than that of the proposed second-order solution (0.3457 s).

Monte Carlo simulations are further performed to investigate the estimation accuracy of the proposed algorithm. Note that the MC simulations performed herein differ from those shown in Figs. 3 and 4, as the MC simulations in Figs. 3 and 4 only examine the accuracy of the modified STT and the second-order measurement model. The state estimated errors of the 100 MC runs (using the second-order solution) of the standard case are shown in Fig. 5. It should be noted that the proposed method can only estimate $\mathbf{x}_0$, $\Delta \mathbf{v}_1$, and t_1 . The state estimated errors (i.e., the blue lines) in Fig. 5 are obtained by propagating the estimations ($\mathbf{x}_0$, $\Delta \mathbf{v}_1$, and t_1) and then comparing them with the true orbits. Moreover, the triple STD (3-STD) (3σ) bounds of the estimated errors, which are calculated numerically using these MC points, are presented by the solid red lines. The position and velocity estimated errors of the 100 MC runs are smaller than 0.75 km and 1.2 m/s, respectively. The distributions of the estimations of $\Delta \mathbf{v}_1$ and t_1 using the second-order solution are presented in Fig. 6. In Fig. 6, the blue and red boxes represent the results of the second-order and linear solutions, respectively, and the circles represent the outliers. Using the second-order solution, the absolute errors of estimations of the impulsive epoch t_1 are no larger than 4 s. The estimation errors of the impulsive vector $\Delta \mathbf{v}_1$ are less than 0.5 m/s per axis. Additionally, the results of the linear solution are shown in Fig. 6 as a comparison. As shown in Fig. 6, the distributions

of the red boxes are larger than those of the blue boxes, indicating that the linear solution has a poor maneuver reconstruction performance.

To investigate the uncertainty quantification (UQ) capacities of the proposed method, the Mahalanobis distances (MDs) are calculated in the MC simulations. The MD is defined as

$$MD = \sqrt{(\hat{\mathbf{X}} - \mathbf{X}_r)^T \hat{\mathbf{P}}^{-1} (\hat{\mathbf{X}} - \mathbf{X}_r)} \quad (71)$$

where $\hat{\mathbf{X}}$ and $\mathbf{X}_r$ are the estimated and true parameter vector (including the initial orbit state and impulsive maneuver epoch and vector), respectively, and $\hat{\mathbf{P}}$ is the covariance matrix predicted by the algorithm. As shown in Eq. (71), $MD = \sqrt{n_X}$ ($n_X = 10$ is the dimension of the parameter vector $\mathbf{X}_r$) indicates that the estimated error $\hat{\mathbf{X}} - \mathbf{X}_r$ lies on the 1σ boundary of the covariance $\hat{\mathbf{P}}$, whereas $MD = 3\sqrt{n_X}$ means that the estimated error lies on the 3σ boundary of the covariance. The MD results of the second-order and linear solutions are plotted in Fig. 7. In Fig. 7, the blue and red boxes represent the MD results of the second-order and linear solutions, respectively, and the circles represent the outliers. Both solutions have MDs smaller than $3\sqrt{n_X}$, and the true estimated errors are inside the 3σ ellipsoid of the covariance predicted by the algorithm. This means that the proposed algorithm [Eq. (66)] can effectively quantify the estimated uncertainties. Additionally, the linear solution has larger MD values, showing that its UQ capacity is poorer than that of the second-order solution [36]. Thus, one can see from Table 4 and Figs. 6 and 7 that the second-order solution outperforms the linear version in terms of convergence, accuracy, and uncertainty quantification capacity. Following in this paper, the second-order solution is employed.

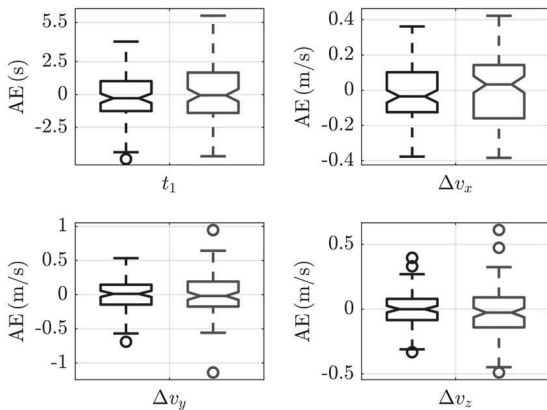


Fig. 6 Estimated error distributions of the impulsive vector and impulsive epoch in the standard case.

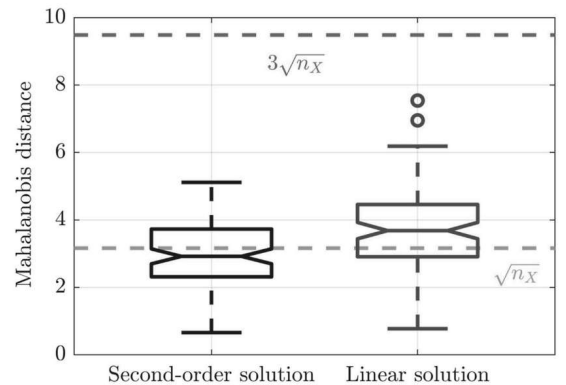


Fig. 7 Mahalanobis distance results of the MC simulation.